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# Effect of Charge Shape and Initiation Configuration of Explosive Cylinders Detonating in Free Air on Blast-Resistant Design

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**Abstract:** This paper investigates the influence of the cylindrical charge characteristics, i.e., the length-to-diameter ratio, orientation, and initiation configuration, on the blast loads (peak overpressure and maximum impulse). Three different initiation configurations were considered, i.e., at the center, at one end, and at both ends. Numerical models were developed for the spherical and cylindrical charges. The numerical results were analyzed along the gauge lines at different azimuth angles around the cylindrical charges. Some important observations were made. Neglecting the charge shape effect, the peak overpressure (maximum impulse) in the near field generated from centrally initiated cylindrical charges can be underestimated by a factor as high as 3.0 (1.9). Therefore, the cylindrical charge shape should be explicitly modeled in the numerical simulations for the blast-resistant design of protective structures subjected to near-field detonations. It is confirmed that the shock front heals into a spherical one in the far field. Hence, the blast loads generated from the spherical charge of the same mass can be used. In addition, the influence range, beyond which the charge shape effect can be neglected, is 5.7 $m/kg^{1/3}$ for the impulse, which is about twice that for the overpressure (3.2 $m/kg^{1/3}$). In general, the maximum values of blast loads resulted by the three considered initiation configurations can be sorted in descending order, i.e., at one end > at the center > at both ends. The initiation configuration at one end (at the center) has the largest influence range of 3.9 $m/kg^{1/3}$ (5.7 $m/kg^{1/3}$) for the overpressure (impulse) among the three considered initiation configurations. **DOI: 10.1061/(ASCE)ST.1943-541X.0002694.** *© 2020 American Society of Civil Engineers.*

**Author keywords:** Charge shape; Length-to-diameter ratio; Initiation configuration; Azimuth angle; Arbitrary Lagrangian-Eulerian.

## Introduction

Modern guidelines, e.g., UFC 3-340-01 (USACE 2002) and UFC 3-340-02 (USACE 2014), are often used during the evaluation process of blast loads on structures. The underlying basis to calculate the blast loads is the empirical formulas of Kingery and Bulmash (1984). This results in two fundamental assumptions being the basis for evolution. First, the charges are assumed to be either spherical (for free air bursts) or hemispherical (for surface bursts). Second, the charges are centrally initiated. This means that the influences of the nonspherical charge shape, charge orientation, and initiation configuration on the blast loads are neglected. To avoid unwanted consequences (e.g., accidental explosions) or to improve the efficiency of the charges (e.g., ammunition destruction), the physical phenomena of the shock wave propagation caused by the nonspherical charges should be analyzed. Thus, the effect of the charge shape, orientation, and initiation configuration on the blast loads can be assessed.

Because most warheads and ordnance devices are cylindrical rather than spherical, the study of shock waves generated from cylindrical charges has practical relevance. Hence, in the past, considerable research has been dedicated to the properties of shock waves generated from cylindrical charges. Stoner and Bleakney (1948) developed empirical formulas to calculate the peak overpressures generated from spherical, cylindrical, and rectangular charges. They indicated that the influence of the charge shape on the peak overpressure is significant until the scaled distance $Z$ is larger than at least 5 $m/kg^{1/3}$. The scaled distance $Z$ is given as $R/W^{1/3}$, where $R$ is the standoff distance from the charge to the gauge and $W$ is the charge mass. Wisotski and Snyer (1965) investigated free air bursts using cylindrical Composition B charges. The length-to-diameter ratio ($L/D$) varies from 1:4 to 10:1 and the mass varies from 0.72 to 3.70 kg. Wisotski and Snyer revealed that there was no indication of the blast loads (peak overpressures and maximum impulses) approaching those generated from spherical charges within the measurement range. Plooster (1982) showed the experimental results of the shock wave properties for free air bursts, in which cylindrical pentolite charges were initiated at one end. The $L/D$ varies from 1:4 to 6:1. The blast parameters were measured around the cylindrical charges with a 22.5° interval of the azimuth angle, which is defined as the angle between the longitudinal axis of the cylindrical charge and the axis of the measurement. Substantial differences were observed in the blast parameters generated from cylindrical and spherical charges. Ismail and Murray (1993) reported blast measurements of cylindrical tetryl charges. Zimmerman et al. (1999) investigated numerically the effect of the charge shape on the overpressures and impulses generated from cylindrical C4 charges. Omang et al. (2009) investigated the effect of the cylindrical charge shape for height-of-burst

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scenarios. Tham (2009) conducted numerical simulations to model the interaction of shock waves with a series of aluminum cylinders in the near field. The numerical results of the momentum transfer to the aluminum cylinders showed good agreement with those obtained from the experiments carried out by Held (1999). Held revealed that the charge shape, the point, and the time of initiation affected the shock wave contour and intensity. Simoens et al. (2012) conducted a series of experiments using centrally initiated cylindrical charges to specify the influence range of the charge shape effect on the overpressures measured along the radial direction. Knock and Davies (2011b, 2013), Knock et al. (2015), and Anastacio and Knock (2016) proposed empirical formulas for the side-on peak overpressure and Knock and Davies (2011a, 2013), Knock et al. (2015), and Anastacio and Knock (2016) proposed maximum impulse generated from cylindrical charges. Recently, Sherkar et al. (2016) conducted numerical studies to characterize the influence of the charge shape and initiation configuration on the side-on overpressures and impulses for free air bursts. They drew two preliminary conclusions. First, the influence of the charge shape on the overpressure (impulse) could be ignored if the scaled distance $Z$ is larger than 3.06 $\mathrm{m/kg^{1/3}}$ (2.11 $\mathrm{m/kg^{1/3}}$). Second, the effect of the initiation configuration (at the center or at one end) on the blast load distribution could be ignored at scaled distances $Z > 3$ $\mathrm{m/kg^{1/3}}$. This information is instructive. However, as commented by Sherkar et al., it was not sufficient to identify the scaled distance beyond which the charge shape effect can be ignored. Additional information is needed, such as the peak overpressures and maximum impulses in the directions other than axial and radial. In addition, cylindrical charges may generate secondary and tertiary shocks in the axial direction that have comparable or even higher peak values than the primary shock (Plooster 1982; Esparza 1986; Ismail and Murray 1993). Sherkar et al. (2016) did not include the increase in the impulse caused by the secondary and tertiary shocks during the calculation of the maximum impulse. Hu et al. (2018) carried out a series of field tests at the close-in range. The cylindrical charges were initiated at both ends. The object was to develop a blast loading model on RC columns that can consider the charge shape and initiation configuration. Hu et al. concluded that the reflected peak overpressure (maximum impulse) generated from the cylindrical charges initiated at both ends is up to 3.5 (2) times larger than those initiated at the center. They attributed the increase in the blast loads to the self-Mach-reflection caused by the initiation at both ends of cylindrical charges at the close-in range.

It was indicated by many previous studies that the shock waves generated from cylindrical charges differ significantly from the ones generated from spherical charges, especially in the near field. The majority of these studies were dedicated to the influence of the charge characteristics on the blast parameters, e.g., length-to-diameter ratio (Stoner and Bleakney 1948; Wisotski and Syner 1965; Swisdak 1975; Plooster and Yatteau 1979; Plooster 1982; Petes 1986; Esparza 1986; Zimmerman et al. 1999; Wu et al. 2010; Simoens et al. 2012; Hryciów et al. 2014; Simoens and Lefebvre 2015; Sherkar et al. 2016), initiation configuration (Ismail and Murray 1993; Schrami et al. 2002; Tham 2009; Wu et al. 2010; Sherkar et al. 2016), and azimuth angle (Wisotski and Syner 1965; Plooster and Yatteau 1979; Guerke and Scheklinski-Glueck 1982; Plooster 1982; Esparza 1986; Zimmerman et al. 1999; Held 2002; Tham 2009; Hryciów et al. 2014). The $L/D$ significantly influences the blast load distribution. More energy is directed in the radial direction for larger $L/D$ ratios, whereas more energy is directed in the axial direction for smaller $L/D$ ratios. The difference in the directed energy between the axial and radial directions increases with the length-to-diameter ratio (Zimmerman et al. 1999; Wu et al. 2010). However, little information (e.g., USDOD 1989) was provided on how the $L/D$ affects the peak overpressures and maximum impulses. USDOD (1989) developed an empirical formula for initial modeling attempts, which relates the peak overpressures ($p_{cyl}$) along the radial direction generated from cylindrical charges to the ones generated from spherical charges ($p_{sph}$), i.e., $p_{cyl}/p_{sph} = 5.53(L/D)^{0.308}/Z^{0.998} + (Z - L/D)/24.99$. The blast loads in the near field depend strongly on how the charge is initiated. Different initiation configurations were considered in the previous studies, i.e., at the center (Schrami et al. 2002; Wu et al. 2010; Simoens et al. 2012; Simoens and Lefebvre 2015; Sherkar et al. 2016), at one end (Wisotski and Syner 1965; Swisdak 1975; Plooster and Yatteau 1979; Guerke and Scheklinski-Glueck 1982; Plooster 1982; Petes 1986; Esparza 1986; Ismail and Murray 1993; Zimmerman et al. 1999; Held 1999; Schrami et al. 2002; Held 2002; Tham 2009; Wu et al. 2010; Maranda et al. 2014; Sherkar et al. 2016; Johnson et al. 2018), and at both ends (Schrami et al. 2002; Tham 2009; Anastacio and Knock 2016; Hu et al. 2018; Chen et al. 2019). The orientation of the cylindrical charge also plays a crucial role in the accurate assessment of the blast parameters (Ismail and Murray 1993). However, discussions about the influence range of the charge characteristics on the blast loads are limited, e.g., Simoens et al. (2012) and Sherkar et al. (2016). In summary, there is still a lack of a systematic analysis of the influence of the charge characteristics on the blast loads, which can be invoked during the blast-resistant design of protective structures.

This paper discusses how the charge characteristics (shape, orientation, and initiation configuration) influence the blast loads generated from cylindrical charges for free air bursts. This work extends the study conducted by Sherkar et al. (2016) in three aspects. First, not only the blast loads in the axial and radial directions but also in other directions around the cylindrical charges are considered. Second, the increase in the impulse caused by secondary and tertiary shocks are also taken into account. Third, the initiation configuration at both ends is also discussed alongside the configuration initiated at the center and at one end. The paper is arranged as follows. The motivations and objectivity of this study were outlined in this introduction section. The numerical models and analyses are described in the next two sections. Based on the numerical results, the influence of the charge characteristics on the peak overpressures and maximum impulses generated from cylindrical charges are investigated. At the end, some conclusions regarding the blast-resistant design of protective structures are summarized.

## Numerical Models

This section describes the numerical models employed to calculate the blast parameters (i.e., peak overpressures and maximum impulses) generated from spherical and cylindrical charges for free air bursts. Numerical simulations were conducted with the software LS-DYNA.

The multimaterial arbitrary Lagrangian-Eulerian (MM-ALE) method is often used to model the process of shock wave propagation. There are two computational burdens in the use of the MM-ALE method. First, a large air domain needs to be modeled to mitigate the boundary effect. Second, the mesh should be sufficiently fine to adequately resolve the shock wave propagation, especially in the vicinity of the charge. According to the Courant-Friedrichs-Lewy (CFL) condition (Courant et al. 1928), this results in a small time step. What is often done in practice is to use a radial mesh in which the element size increases with the distance from the charge (Schwer 2016). In addition, Rigby (2014) demonstrated that

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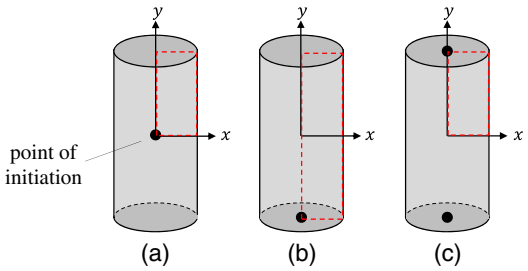



**Fig. 1.** Initiation configurations: (a) center-initiated; (b) single-end-initiated; and (c) double-end-initiated.

the radial mesh can better simulate the shock wave propagation because the elements are generally oriented in the direction of the shock wave propagation, which reduces the second-order advection errors and results in less numerical dispersion. The radial mesh has already been successfully employed by Xiao et al. (2019) to derive the trinitrotoluol (TNT) equivalence factor of the high explosive material pentaerythrityltetranitrat (PETN). Hence, it is adopted in this paper to conduct the numerical simulations for cylindrical charges.

Three different initiation configurations (Fig. 1) were considered. They are denoted as center-initiated (CI), single-end-initiated (SEI), and double-end-initiated (DEI). This means that the charge is detonated at the center, at one end, and at both ends of the cylinder, respectively.

For spherical charges, one-dimensional (1D) MM-ALE simulation was run with spherical element formulation. The 1D model [Fig. 2(a)] had a length of 24 m. For cylindrical charges, two-dimensional (2D) axisymmetric MM-ALE simulations were run. If the cylindrical charge was initiated at the center [Fig. 1(a)] or at both ends [Fig. 1(c)], the pertinent 2D numerical model was composed of a quarter circle [Fig. 2(b)]. The $x$- and $y$-axes were the axis of the mirror and rotational symmetry, respectively. Thus,

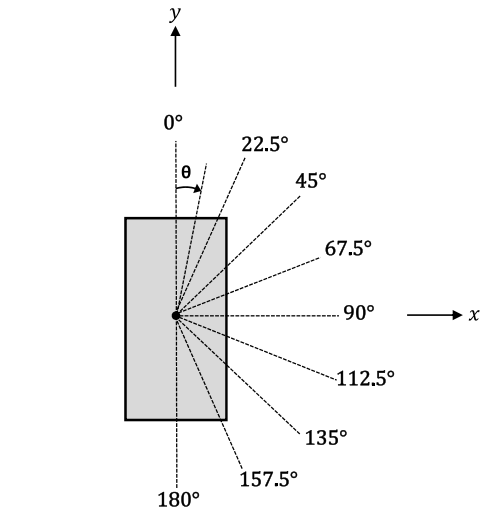



**Fig. 3.** Definition of the azimuth angles of cylindrical charges.

the computational time can be reduced. If the cylindrical charge was initiated at one end [Fig. 1(b)], the mirror symmetry was no longer valid. Hence, the pertinent 2D numerical model consisted of a half circle [Fig. 2(c)]. The $y$-axis was the axis of rotational symmetry. The radius of the 2D numerical models was 24 m.

Different definitions of azimuth angle are present in previous publications. The following definition (Fig. 3) is used in this paper: The azimuth angle is 0° at the top end of the cylinder, 90° when perpendicular to the axis of the cylinder, and 180° when at the bottom end of the cylinder. In an attempt to examine the blast load distribution around the cylindrical charges, 9 (17) gauge lines along the azimuth angle $\theta$ varying from 0° to 90° (180°) with an interval of $\bar{\theta} = 11.25°$ were considered for the 2D numerical model illustrated in Figs. 2(b and c).

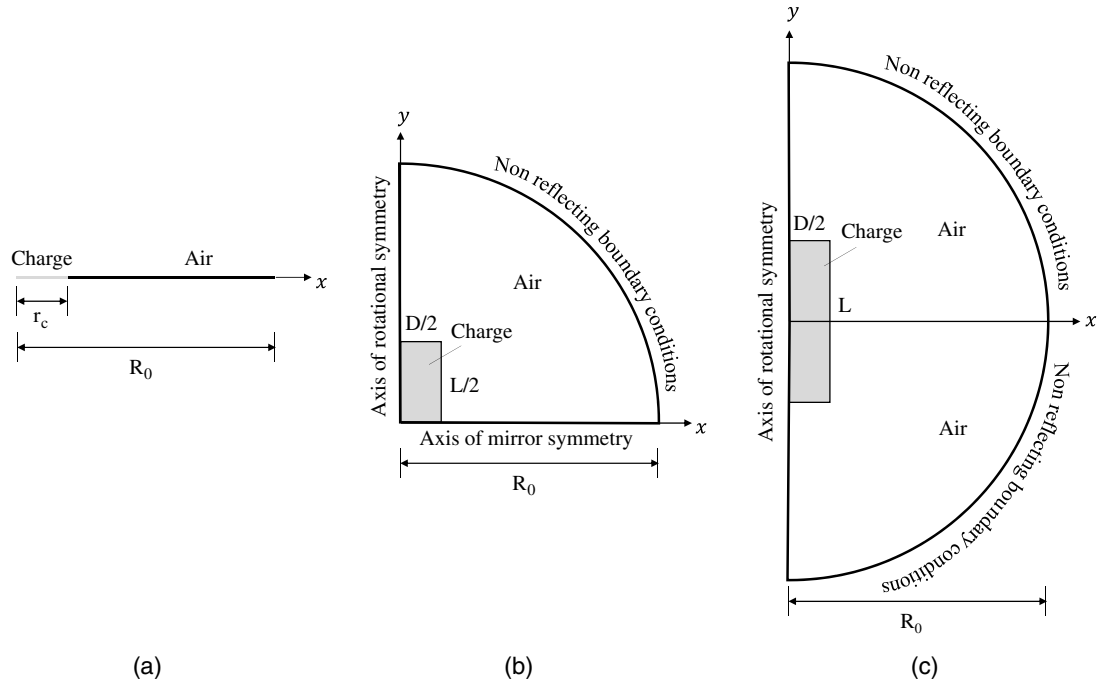



**Fig. 2.** Numerical models: (a) 1D, center-initiated spherical charges; (b) 2D, center-initiated or double-end-initiated cylindrical charges; and (c) single-end-initiated cylindrical charges.

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**Table 1.** Dimensions of cylindrical charges, 8 kg TNT

| $L/D$ (—) | $L$ (cm) | $D$ (cm) |
|---|---|---|
| 1 | 18.42 | 18.42 |
| 2 | 29.24 | 14.62 |
| 3 | 38.31 | 12.77 |
| 4 | 46.41 | 11.60 |
| 5 | 53.86 | 10.77 |

For the sake of simple calculation for the scaled distance, a charge of 8 kg TNT was used in the numerical simulations. The distance from the charge center to the gauges varied from 1 to 20 m with an interval of 0.2 m. This corresponds to a scaled distance $Z$ from 0.5 to 10 m/kg$^{1/3}$ with an interval of 0.1 m/kg$^{1/3}$. Based on the overpressure–time histories captured at the gauges, the peak overpressures and maximum impulses can be calculated. The spherical charge had a radius of 10.54 cm. The dimensions of the cylindrical charges are given in Table 1. Five different $L/D$s, varying from 1 to 5, were included in the numerical analyses. The spherical and cylindrical shape of the charges was modeled via the keyword *INITIAL_VOLUME_FRACTION_GEOMETRY.

### Strength Model and Equation of State

The explosive charge and air domain were explicitly modeled. The explosive charge was modeled via *MAT_HIGH_EXPLOSIVE_BURN with the Jones-Wilkins-Lee (JWL) equation of state (EoS). As assumed in Sherkar et al. (2016), the effect of afterburning was neglected in this study. The nonuniform afterburning has little influence on the peak overpressure but can cause an increase in the maximum impulse (Shin et al. 2014).

Typical material parameters of the strength model and EoS for TNT are listed in Table 2, where $\rho_e$ is the density of the explosive material, $D$ is the velocity of detonation, $p_{CJ}$ is the Chapman-Jouguet pressure, $e_0$ is the specific internal energy, and $A$, $B$, $R_1$, $R_2$, and $\omega$ are material constants.

The air was modeled via *MAT_NULL and regarded as an ideal gas with a linear polynomial EoS. The standard atmospheric conditions were assumed in the numerical analyses, i.e., at mean sea level and at the standard temperature of 15°C, which result in an atmospheric pressure of 101.325 kPa and an air density of 1.225 kg/m$^3$ (Xiao et al. 2018).

### Mesh Refinement Study

To determine a reasonable element size for the numerical models for cylindrical charges, a mesh refinement study was conducted using five different meshes. The nominal element sizes in the vicinity of the charges were 4, 2, 1, 0.5, and 0.25 mm, respectively. The numerical model of the center-initiated cylindrical charges [Fig. 2(b)] was used to conduct the study. The numerical simulations were terminated at 55 ms when the shock wave had passed the furthest gauge located 20 m from the charge. A workstation including 64 processors was employed to carry out the simulations. The total number of elements in the numerical models and the computational time required to run each simulation are given in Table 3.

**Table 2.** Material parameters of strength model and EoS for TNT

| Parameters | Values |
|---|---|
| Explosive | TNT |
| $\rho_e$ (kg/m$^3$) | 1,630 |
| $D$ (m/s) | 6,930 |
| $p_{CJ}$ (Pa) | $2.10 \times 10^{10}$ |
| $A$ (Pa) | $3.712 \times 10^{11}$ |
| $B$ (Pa) | $3.231 \times 10^{9}$ |
| $R_1$ (—) | 4.15 |
| $R_2$ (—) | 0.95 |
| $\omega$ (—) | 0.30 |
| $e_0$ (Pa) | $7.00 \times 10^{9}$ |

Source: Data from Dobratz and Crawford (1985).

**Table 3.** Number of elements and computational time in the mesh refinement study

| Features of numerical simulations | Nominal element size (mm) | | | | |
|---|---|---|---|---|---|
| | 4.0 | 2.0 | 1.0 | 0.5 | 0.25 |
| Number of elements (—) | 7,203 | 28,812 | 115,287 | 481,068 | 1,824,672 |
| Computational time (h) | 0.1 | 0.2 | 0.4 | 2.3 | 15.2 |

Fig. 4 shows the overpressure–time histories at four gauges along the gauge line $\theta = 45°$ of the cylindrical charges, which were obtained from the numerical simulations using the five different nominal element sizes mentioned previously. The scaled distances from the charge to the gauges were 0.5, 2.5, 5, and 10 m/kg$^{1/3}$, respectively. The $L/D$ was 2. It is observed that the analyses using nominal element sizes of 0.25 and 0.5 mm produce essentially identical results.

To evaluate, if the convergence and fidelity of the numerical model were achieved, the peak overpressures and maximum impulses were selected as quantities for comparison. They were calculated from the overpressure–time histories shown in Fig. 4. Fig. 5 depicts the blast parameters (peak overpressure and maximum impulses) versus the nominal element size. It is observed that the element size has a more significant influence on the peak overpressure than on the maximum impulse. The peak overpressures and maximum impulses at all gauges converge as the nominal element size decreases. Therefore, the finest mesh (nominal element size of 0.25 mm) was selected to carry out the numerical simulations in the following sections. The element size in the vicinity of the charge was 0.25 mm, whereas the element size for the air domain radially increased from 0.25 mm to 5.8 cm in the direction of the shock wave propagation. Thus, the 2D numerical model illustrated in Figs. 2(b and c) has in total 1.8 (3.6) million elements.

The 1D numerical model [Fig. 2(a)] was discretized using the element size of 0.25 mm based on three considerations. First, it is the same as the nominal element size of the 2D numerical models for cylindrical charges. Second, a sufficiently small element size is important to accurately model the explosive energy release. As a rule of thumb, at least 10 elements across the charge radius should be used to mesh the explosive charge (Schwer and Rigby 2018). The selected element size (0.25 mm) is much finer than the one according to the rule of thumb. It provides 421 elements across the radius of the spherical charge. Third, the selected element size (0.25 mm) is smaller than that recommended by Shin et al. (2014), who concluded that an element size of 0.002 times the distance from the charge to the measuring location was expected to provide converged solutions for the numerical modeling of close-in detonations. In addition, it agrees with the element size used by Shin et al. (2015). The authors performed thorough mesh refinement studies for the incident and normally reflected overpressure and impulse, which are generated from detonations of spherical high explosives in free air. Because of these reasons, a mesh refinement study is omitted for the 1D numerical model. Thus, the 1D model has in total 96,000 elements.

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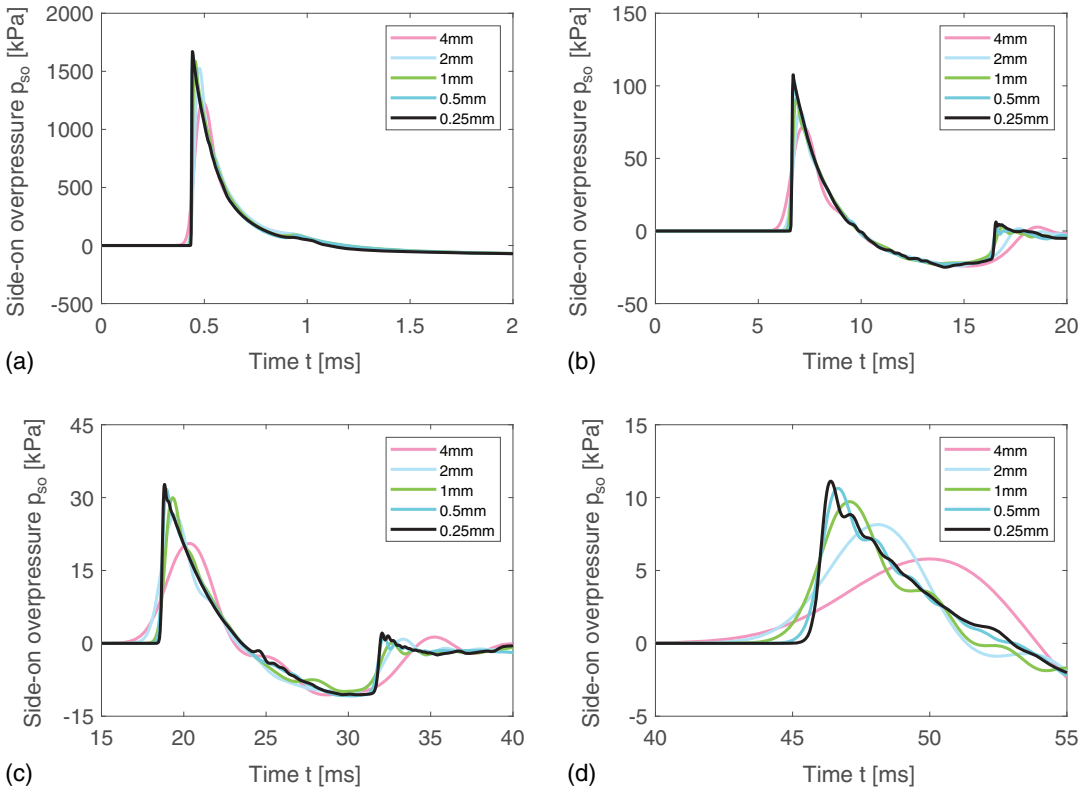



**Fig. 4.** Overpressure–time histories at the gauges along the gauge line $\theta = 45°$, 8 kg cylindrical TNT charges, center-initiated, and $L/D = 2$: (a) $Z = 0.5$ m/kg$^{1/3}$; (b) $Z = 2.5$ m/kg$^{1/3}$; (c) $Z = 5$ m/kg$^{1/3}$; and (d) $Z = 10$ m/kg$^{1/3}$.

## Numerical Analyses

For cylindrical charges, the resulting shock wave patterns are quite complex [see, e.g., Fig. 1 in Sherkar et al. (2016)]. At first, spherical waves are generated in the vicinity of the cylindrical charge along the radial and axial directions, which are denoted as primary side and end waves. After a certain time, these waves interact with each other as they propagate away from the charge. As a result, a reinforced nonspherical wave is formed, which is referred to as the bridge wave (Ismail and Murray 1993; Johnson et al. 2018). Finally, the shock front heals into a spherical one in the far field.

To correlate the blast parameters generated from cylindrical charges with the ones generated from spherical charges, the shape factors $S_p$ and $S_i$ for the overpressure and impulse are introduced. They are defined as follows:

$$S_p = \frac{p_{cyl}}{p_{sph}} \tag{1}$$

$$S_i = \frac{i_{cyl}}{i_{sph}} \tag{2}$$

where $p_{cyl}$ and $i_{cyl}$ ($p_{sph}$ and $i_{sph}$) = peak overpressure and maximum impulse generated from cylindrical (spherical) charges. Regarding a centrally initiated spherical charge in free air, the explosive energy release is regarded as proportional in all directions. Hence, a spherical shock wave propagation can be assumed. Given a specified explosive material, the resulting blast parameters depend only on the charge mass $W$ and standoff distance $R$, i.e., $p_{sph} = f(W, R)$ and $i_{sph} = g(W, R)$. The relationship can be further simplified by using the scaled distance $Z$, i.e., $p_{sph} = f(Z)$ and $i_{sph} = g(Z)$. An example is the empirical formulas of Kingery and Bulmash (1984). In addition to the scaled distance $Z$, the blast parameters generated from cylindrical charges also depend on the length-to-diameter ratio $L/D$, azimuth angle $\theta$, and initiation configuration (IC), i.e., $p_{cyl} = f(Z, L/D, \theta, \text{IC})$ and $i_{cyl} = g(Z, L/D, \theta, \text{IC})$. Accordingly, the shape factors $S_p$ and $S_i$ for the overpressure and impulse depend also on these quantities, i.e., $S_p = f(Z, L/D, \theta, \text{IC})$ and $S_i = g(Z, L/D, \theta, \text{IC})$.

### *Length-to-Diameter Ratio*

This section explores the influence of $L/D$ on the blast parameters generated from cylindrical charges. The $L/D$ value varies from 1 to 5 and the cylindrical charges are initiated at the center.

Fig. 6 compares the side-on overpressure– and impulse–time histories generated from cylindrical charges with $L/D = 2$ and 4 along the axial and radial directions with the ones generated from spherical charges. It is observed that multiple shocks, which are less significant than the primary shock, are seen in the overpressure–time histories at small scaled distances along the axial direction, e.g., $Z = 1.8$ m/kg$^{1/3}$ [Fig. 6(a)]. This leads to an increase in the maximum impulse [Fig. 6(c)]. Secondary shocks, which are of larger magnitudes than the primary shocks, are present in the overpressure–time histories at large scaled distances along the axial direction, e.g., $Z = 5.7$ m/kg$^{1/3}$ [Fig. 6(b)]. This significantly increases the maximum impulse along the axial direction. The overpressure–time histories along the radial direction exhibit a form similar to the ones generated from spherical charges. This is most likely related to the spherical curvature in the radial direction of cylindrical charges.

The overpressure and impulse contours generated from cylindrical charges versus the azimuth angle are shown in Fig. 7. The overpressure and impulse contours are depicted by the shape

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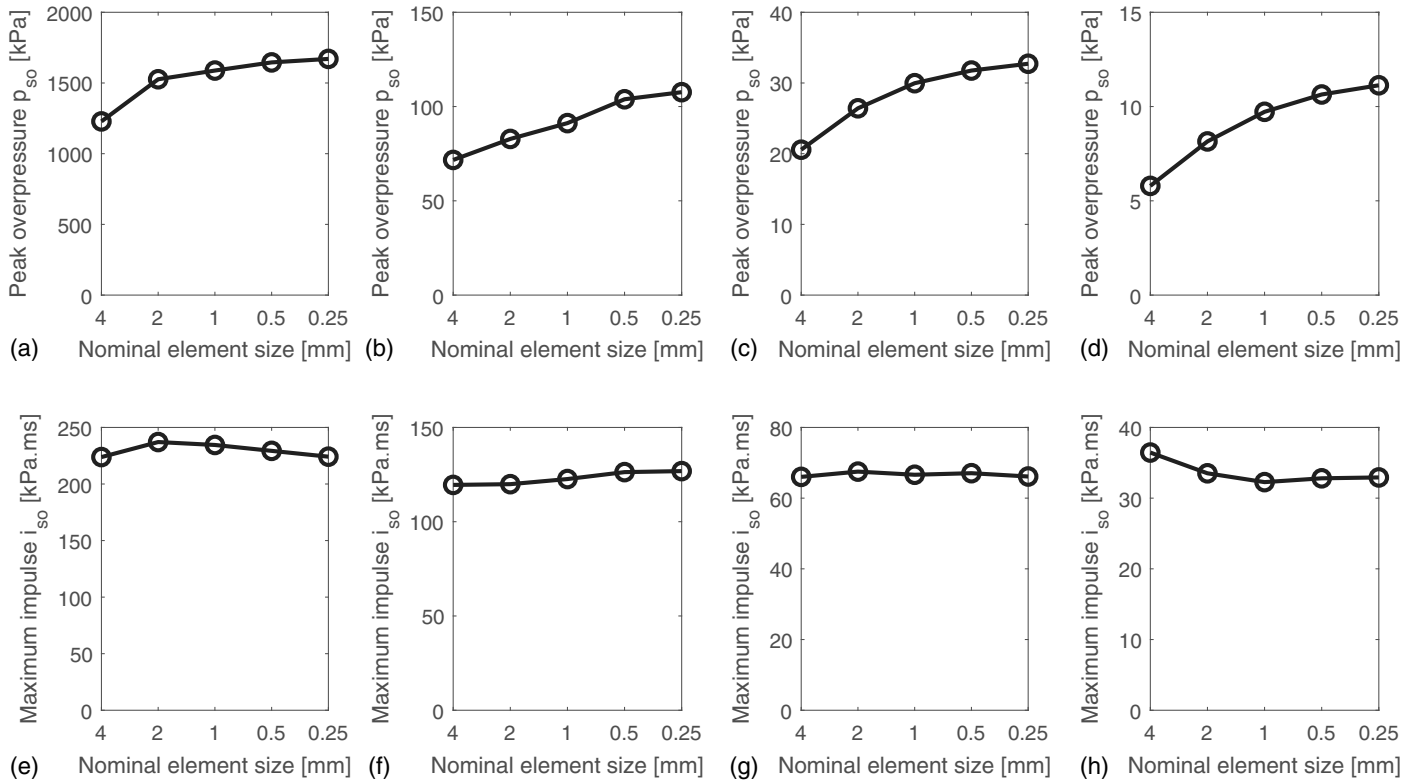



**Fig. 5.** Blast parameters at the gauges along the gauge line $\theta = 45°$, 8 kg cylindrical TNT charges, center-initiated, and $L/D = 2$: (a) peak overpressure, $Z = 0.5$ m/kg$^{1/3}$; (b) peak overpressure, $Z = 2.5$ m/kg$^{1/3}$; (c) peak overpressure, $Z = 5$ m/kg$^{1/3}$; (d) peak overpressure, $Z = 10$ m/kg$^{1/3}$; (e) maximum impulse, $Z = 0.5$ m/kg$^{1/3}$; (f) maximum impulse, $Z = 2.5$ m/kg$^{1/3}$; (g) maximum impulse, $Z = 5$ m/kg$^{1/3}$; and (h) maximum impulse, $Z = 10$ m/kg$^{1/3}$.

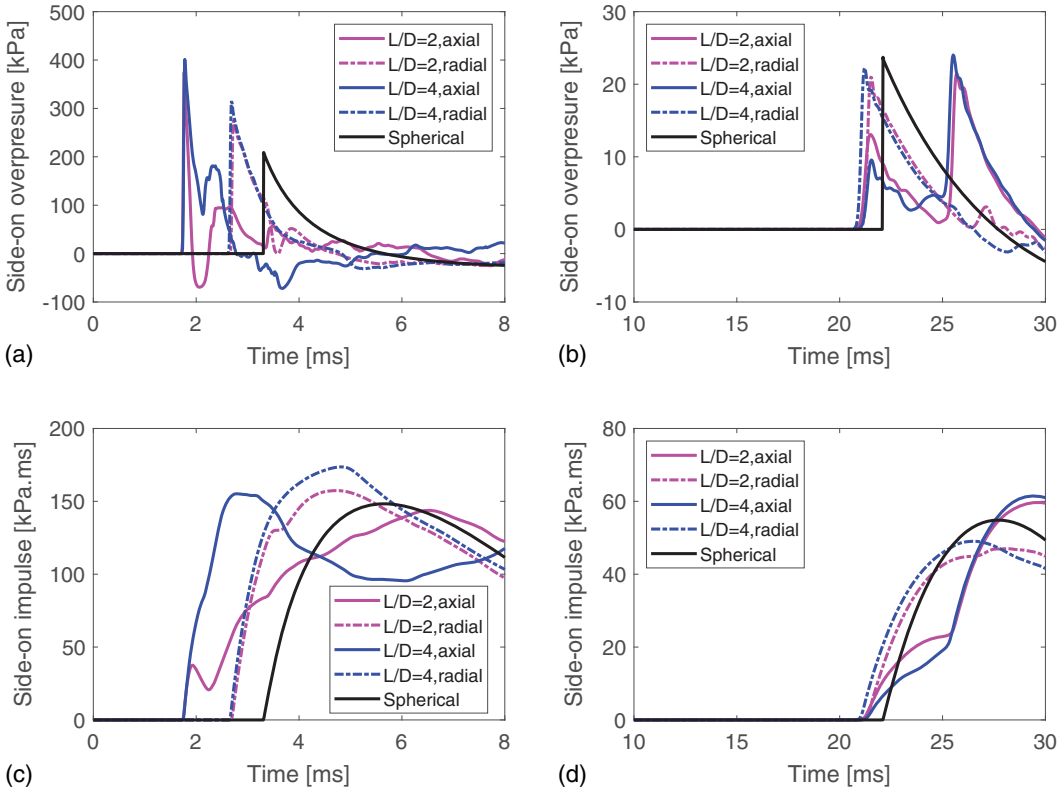



**Fig. 6.** Side-on overpressure– and impulse–time histories generated from cylindrical and spherical charges, center-initiated: (a) overpressure, $Z = 1.8$ m/kg$^{1/3}$; (b) overpressure, $Z = 5.7$ m/kg$^{1/3}$; (c) impulse, $Z = 1.8$ m/kg$^{1/3}$; and (d) impulse, $Z = 5.7$ m/kg$^{1/3}$.



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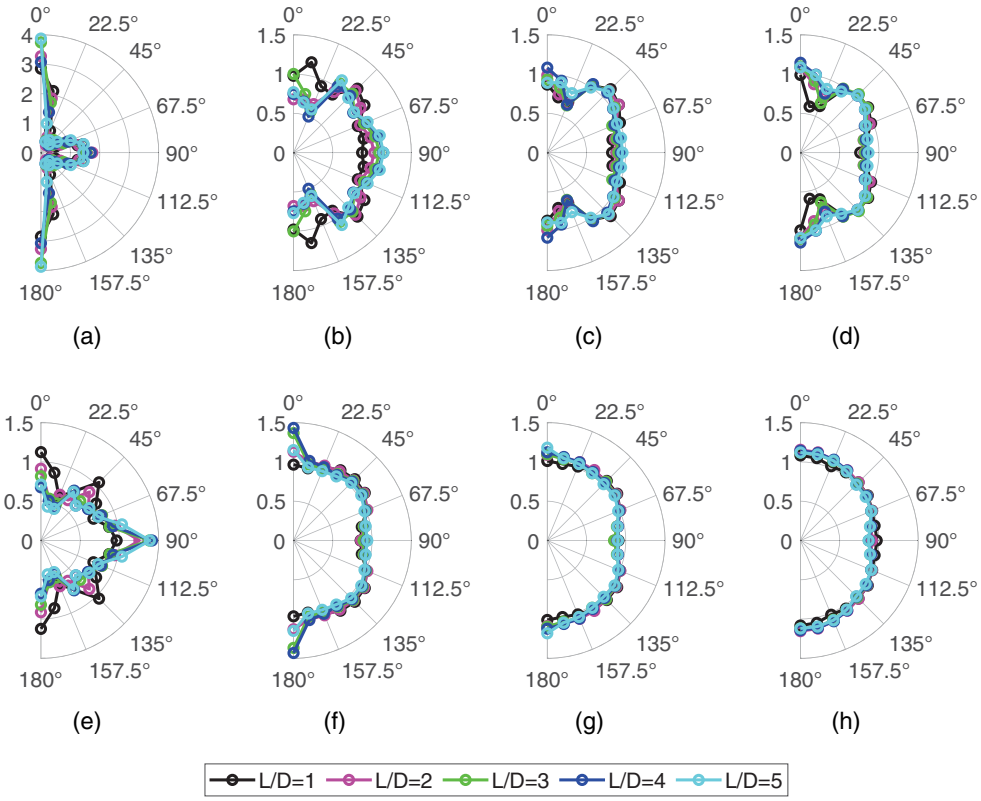



**Fig. 7.** Shape factors ($S_p$ and $S_i$) for the overpressure and impulse versus the azimuth angle for different $L/D$ values, center-initiated: (a) $S_p$, $Z = 0.5$ m/kg$^{1/3}$; (b) $S_p$, $Z = 3.0$ m/kg$^{1/3}$; (c) $S_p$, $Z = 5.7$ m/kg$^{1/3}$; (d) $S_p$, $Z = 10.0$ m/kg$^{1/3}$; (e) $S_i$, $Z = 0.5$ m/kg$^{1/3}$; (f) $S_i$, $Z = 4.0$ m/kg$^{1/3}$; (g) $S_i$, $Z = 5.7$ m/kg$^{1/3}$; and (h) $S_i$, $Z = 10.0$ m/kg$^{1/3}$.

factors ($S_p$ and $S_i$) for the overpressure and impulse at the gauges of the same distance around the charge.

Initiating the cylindrical charges at the center produces symmetric overpressure and impulse contours. However, they are far from spherical at small scaled distances, e.g., $Z = 0.5$ m/kg$^{1/3}$ [Figs. 7(a and e)]. The overpressure contours expand much more significantly along the axial direction than the radial direction because the shock wave propagates more quickly along the axial direction than the radial direction at small scaled distances [Fig. 6(a)]. Special effects can be observed for the impulse contours at small scaled distances. As the $L/D$ value increases, the impulse contours expand along the radial direction and contract along the axial direction [Fig. 7(e)]. This is attributed to the fact that the energy directed along the radial direction increases with $L/D$, whereas the energy directed along the axial direction decreases with $L/D$. As the scaled distance increases, e.g., at $Z = 3.0$ m/kg$^{1/3}$ [Fig. 7(b)], the overpressure contour exhibits a similar four-leafed rose shape (only half is shown due to the rotational symmetry). This means that the contours nominally follow a $\cos(2\theta)$ pattern, i.e., the maximums occur at 0°, 90°, and 180°, and minimums at intermediate angles. Roughly speaking, the overpressure contour tends to be spherical as the scaled distance increases [Figs. 7(b–d)]. The impulse contour at $Z = 4.0$ m/kg$^{1/3}$ [Fig. 7(f)] is nearly spherical except at $\theta = 0°$ and 180°, where the shape factors $S_i$ for the impulse are somewhat larger than the ones at the other angles. As the scaled distance further increases, e.g., $Z = 5.7$ and 10.0 m/kg$^{1/3}$ [Figs. 7(g and h)], the impulse contour can be considered spherical.

Fig. 8 shows the shape factors ($S_p$ and $S_i$) for the overpressure and impulse along the axial and radial directions versus the scaled distance $Z$. Because of the presence of secondary shocks, no valid statement can be made for the influence of $L/D$ on the shape factor $S_p$ for the overpressure along the axial direction [Fig. 8(a)]. Generally, the shape factor $S_i$ for the impulse along the axial direction decreases with $L/D$ at small scaled distances [Fig. 8(b)]. This is because less blast energy is directed along the axial direction as $L/D$ increases. In Fig. 8(c), the shape factor $S_p$ for the overpressure along the radial direction increases with $L/D$ because more blast energy is directed along the radial direction if $L/D$ increases. The same trend can be observed in the shape factor $S_i$ for the impulse along the radial direction at small scaled distances [Fig. 8(d)], whereas the opposite effect is observed at large scaled distances. This is most likely attributed to the trend that the higher the value of $L/D$, the shorter the duration of the positive phase.

In general, the peak overpressure versus the scaled distance can be divided into three different regions [Figs. 8(a and c)]. At small scaled distances (e.g., $Z < 2.0$ m/kg$^{1/3}$ for cylindrical charges with $L/D = 4$), the peak overpressures along the axial direction are larger than those along the radial direction because the shock wave propagates faster along the axial direction than the radial direction. To obtain a pressure balance, the air flows from the high-pressure region (axial) to the low-pressure region (radial). After a certain time, the shock wave propagates slightly faster along the radial direction than the axial direction [Fig. 6(b)]. This gives rise to the larger peak overpressures along the radial direction than the axial direction at intermediate scaled distances (e.g., $Z = 2.0$–$5.1$ m/kg$^{1/3}$ for cylindrical charges with $L/D = 4$). At large scaled distances (e.g., $Z > 5.1$ m/kg$^{1/3}$ for cylindrical charges with $L/D = 4$), the peak overpressures along the axial direction again become larger than those along the radial direction. This is most likely attributed to the secondary shocks caused by the bridge waves. These secondary shocks

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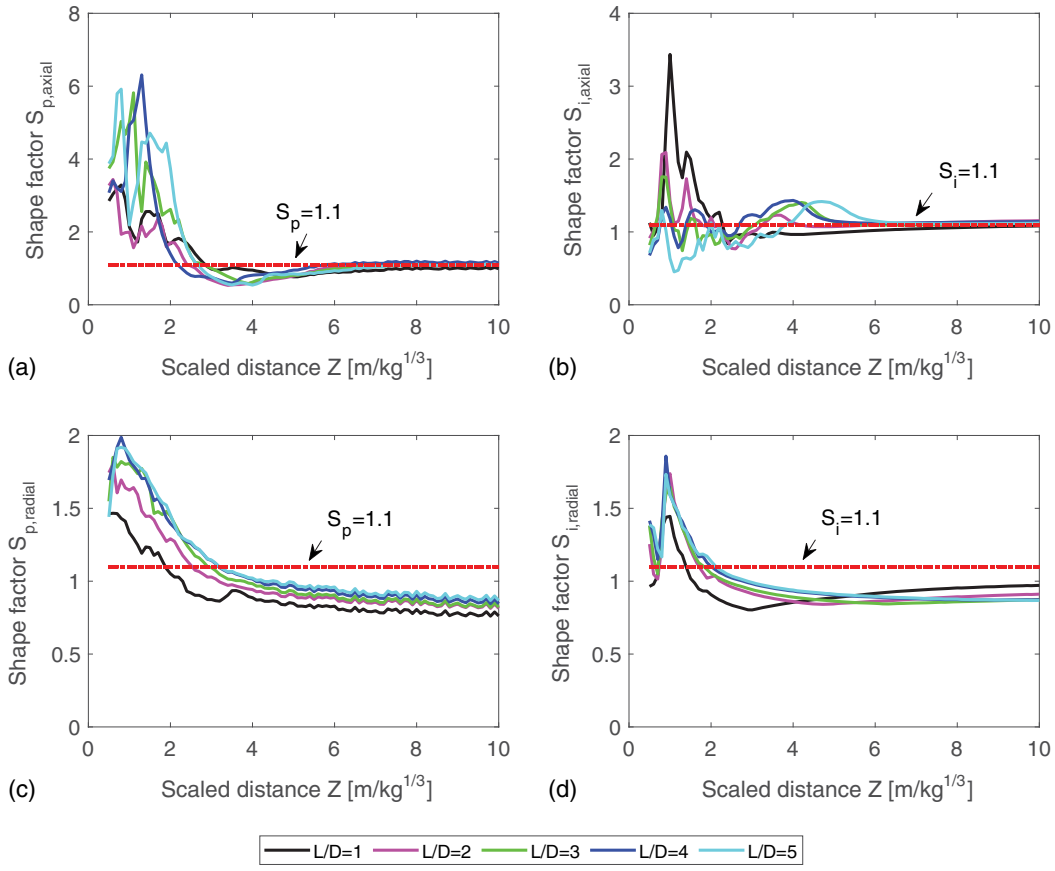



**Fig. 8.** Shape factors for overpressure and impulse versus scaled distance for different $L/D$ values, center-initiated: (a) $S_{p,\text{axial}}$; (b) $S_{i,\text{axial}}$; (c) $S_{p,\text{radial}}$; and (d) $S_{i,\text{radial}}$.

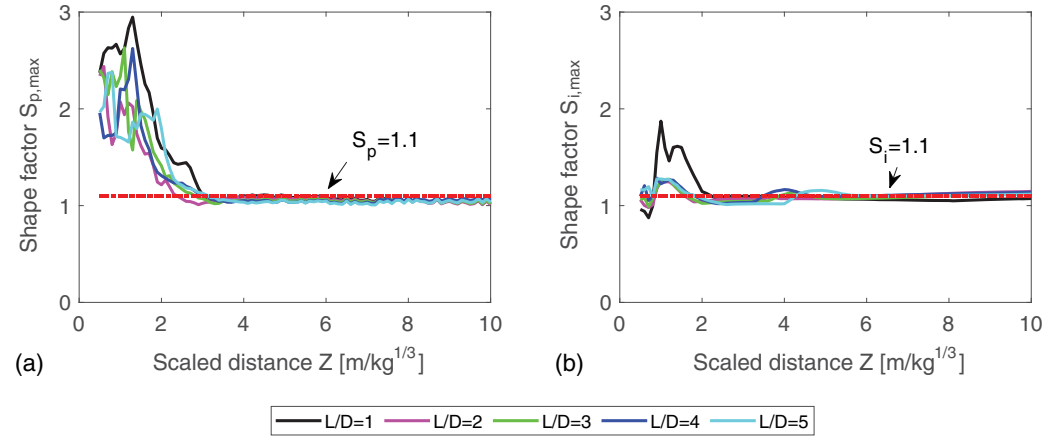



**Fig. 9.** Maximum shape factors for the overpressure and impulse versus the scaled distance for different $L/D$ values, center-initiated: (a) $S_{p,\text{max}}$; and (b) $S_{i,\text{max}}$.

are of larger magnitudes than the primary shock. For cylindrical charges with $L/D < 3$, the maximum impulses are larger along the axial direction than the radial direction at small and large scaled distances, while the opposite effect is observed at intermediate scaled distances. For more elongated cylindrical charges with $L/D \geq 3$, the maximum impulses at small scaled distances (e.g., $Z < 3.0$ m/kg$^{1/3}$ for cylindrical charges with $L/D = 4$) along the radial direction are larger than those along the axial direction because more energy is directed along the radial direction for more elongated charges. At large scaled distances (e.g., $Z > 3.0$ m/kg$^{1/3}$ for cylindrical charges with $L/D = 4$), the maximum impulses are generally larger along the axial direction than the radial direction. This is attributed to the significant increase in the impulse contributed by the secondary and tertiary shocks.

Sherkar et al. (2016) stated that the charge shape effect could be neglected if the peak overpressure and maximum impulse





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generated from the cylindrical charge fall below 110% of the ones generated from the spherical charge. This statement was based on two considerations. First, the overpressure–time histories along the axial direction deviate significantly from the traditional ones used for design, i.e., the triangular or Friedlander curve. Second, it is assumed that a difference of 10% has no significant consequence on the blast-resistant design. This corresponds to the critical values of the shape factors of $S_p = 1.1$ and $S_i = 1.1$, which are defined by Eqs. (1) and (2). They are marked as horizontal dashed lines in Fig. 8. Thus, the scaled distance range, beyond which the charge shape effect is negligible, can be identified. It is 3.2 m/kg$^{1/3}$ for the peak overpressure, which agrees well with the result from Sherkar et al. (2016) (3.06 m/kg$^{1/3}$). However, it is 6.0 m/kg$^{1/3}$ for the maximum impulse, which is much larger than the one from Sherkar et al. (2016) (2.11 m/kg$^{1/3}$). This is because Sherkar et al. (2016) did not account for the increase in the impulse caused by the secondary and tertiary shocks, whereas such impulse increase is considered in this paper.

In addition to the information along the axial and radial directions, information along the other directions is needed to identify the scaled distance range, beyond which the charge shape effect can be neglected. Considering that a target structure is likely to experience loadings associated with a range of azimuth angles, the shape factors $S_p$ and $S_i$ are averaged over a certain ray angle, say, 33.75° (three times the interval of the azimuth angle). For instance, the averaged shape factor $S_{p,\text{ave}}$ ($S_{i,\text{ave}}$) at $\theta = 45°$ is the mean value of the shape factors $S_p$ ($S_i$) at angles $\theta = 33.75°$, $\theta = 45°$, and $\theta = 56.25°$. Next, the maximum shape factors ($S_{p,\max}$ and $S_{i,\max}$) for the overpressure and impulse are calculated as the maximum value among the averaged shape factors for all considered azimuth angles around the cylindrical charge

$$S_{p,\max} = \max\{S_{p,\text{ave}}(Z, L/D, \theta, \text{IC})\} = f(Z, L/D, \text{IC}) \quad (3)$$

$$S_{i,\max} = \max\{S_{i,\text{ave}}(Z, L/D, \theta, \text{IC})\} = g(Z, L/D, \text{IC}) \quad (4)$$

Fig. 9 illustrates the maximum shape factors ($S_{p,\max}$ and $S_{i,\max}$) for the overpressure and impulse versus the scaled distance $Z$. In the near field, the peak overpressure (maximum impulse) generated from the cylindrical charge can be as high as 3.0 (1.9) times the ones generated from the spherical charge. As postulated by many researchers, the blast parameters generated from nonspherical charges smoothen to the ones generated from spherical charges in the far field. Indeed, the shock front heals into a spherical one in the far field.

Considerable uncertainties are involved in a variety of influencing factors of blast loads, such as the type of explosive, actual charge shape, standoff distance, charge orientation with respect to the target structure, atmospheric conditions, and shielding in the far field. Furthermore, the component capacity calculations are mainly conducted in an approximate or simplified manner. Under such circumstances, the previously mentioned statement by Sherkar et al. (2016) was also employed in this study to define the influence range, beyond which the charge shape effect can be neglected. The maximum shape factors ($S_{p,\max}$ and $S_{i,\max}$) do not always occur at the gauges along the axial or radial direction ($\theta = 0°$ or 90°), e.g., for the cylindrical charge with $L/D = 1$, the maximum shape factor $S_{p,\max}$ ($S_{i,\max}$) occurs at 45° in the scaled distance range between 3.7 and 6.9 m/kg$^{1/3}$ (between 2.3 and 8.0 m/kg$^{1/3}$). This is attributed to the presence of the bridge wave. Roughly speaking, the $L/D$ value does not significantly affect the influence range $Z_p$ for the overpressure [Fig. 9(a)], whereas the influence range $Z_i$ for the impulse is seen to expand as $L/D$ increases [Fig. 9(b)]. The largest

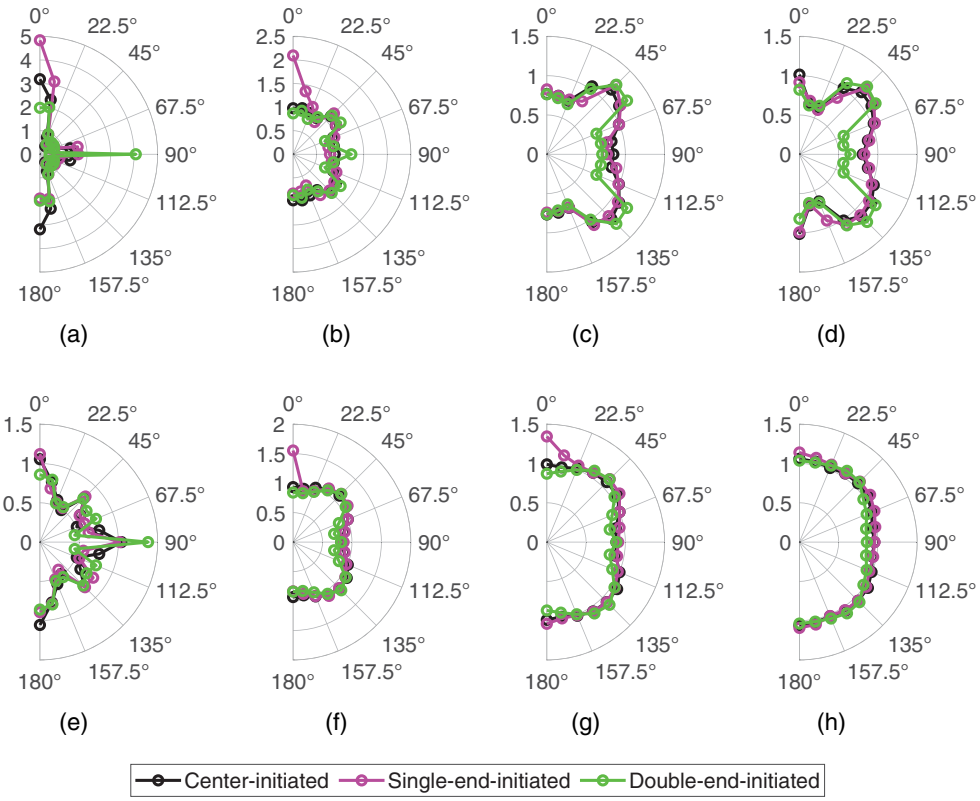



**Fig. 10.** Shape factors ($S_p$ and $S_i$) for the overpressure and impulse dependent on the initiation configuration, $L/D = 1$: (a) $S_p$, $Z = 0.7$ m/kg$^{1/3}$; (b) $S_p$, $Z = 3.3$ m/kg$^{1/3}$; (c) $S_p$, $Z = 5.0$ m/kg$^{1/3}$; (d) $S_p$, $Z = 8.0$ m/kg$^{1/3}$; (e) $S_i$, $Z = 0.7$ m/kg$^{1/3}$; (f) $S_i$, $Z = 3.3$ m/kg$^{1/3}$; (g) $S_i$, $Z = 5.0$ m/kg$^{1/3}$; and (h) $S_i$, $Z = 8.0$ m/kg$^{1/3}$.



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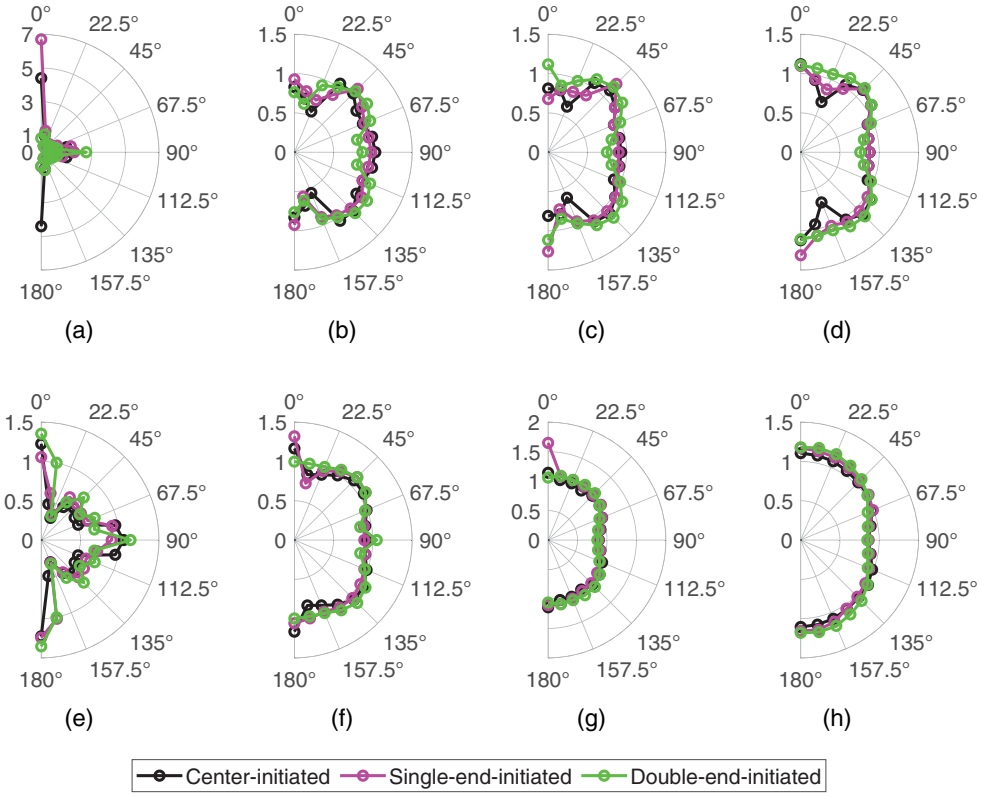



**Fig. 11.** Shape factors ($S_p$ and $S_i$) for the overpressure and impulse dependent on the initiation configuration, $L/D = 3$: (a) $S_p$, $Z = 0.7$ m/kg$^{1/3}$; (b) $S_p$, $Z = 3.3$ m/kg$^{1/3}$; (c) $S_p$, $Z = 5.0$ m/kg$^{1/3}$; (d) $S_p$, $Z = 8.0$ m/kg$^{1/3}$; (e) $S_i$, $Z = 0.7$ m/kg$^{1/3}$; (f) $S_i$, $Z = 3.3$ m/kg$^{1/3}$; (g) $S_i$, $Z = 5.0$ m/kg$^{1/3}$; and (h) $S_i$, $Z = 8.0$ m/kg$^{1/3}$.

influence range for the impulses is 5.7 m/kg$^{1/3}$, which is nearly two times the one for the overpressure (3.2 m/kg$^{1/3}$). Regarding the importance of $L/D$, these findings confirm similar outcomes observed by Sherkar et al. (2016).

### *Initiation Configurations*

Because of the time and directional effects, the center-initiated cylindrical charges may generate much higher blast loads than the center-initiated spherical charges of the same mass. For a center-initiated spherical charge, the detonation wave reaches all the points on the charge surface simultaneously, and shortly thereafter a spherical shock front is formed. Whether a cylindrical charge is initiated at the center, at one end, or at both ends, the sweeping detonation wave will cause the shock front to be nonspherical and to develop a nonspherical volume surrounding the charge. Hence, it is meaningful to study the influence of the initiation configurations on the blast parameters generated from cylindrical charges.

Figs. 10 and 11 illustrate the shape factors ($S_p$ and $S_i$) for the overpressure and impulse that depend on the initiation configurations, i.e., center-initiated, single-end-initiated, and double-end-initiated, for $L/D = 1$ and 3, respectively. As mentioned in the previous section, symmetric overpressure and impulse contours are generated if the cylindrical charge is initiated at the center. Unlike the initiation configuration at the center, the overpressure and impulse contours are skewed in the direction of the detonation wave ($\theta = 0°$) for the initiation configuration at the end of $\theta = 180°$. A highly nonuniform overpressure distribution is observed at small scaled distances [e.g., $Z = 0.7$ m/kg$^{1/3}$ in Figs. 10(a) and 11(a)]. The shock wave is focused in the axial direction. Furthermore, the peak overpressures are substantially larger at the side opposite to the initiation ($\theta = 0°$) than the initiation side ($\theta = 180°$). If the cylindrical charge is initiated at both ends, the overpressure and impulse contours are symmetric. At small scaled distances, the overpressure contour shows an arrow shape, which is oriented along the radial direction [Figs. 10(a) and 11(a)]. This means that, compared to the initiation configuration at the center, the resulting shock wave is focused along the radial direction. However, these highly directional effects caused by different initiation configurations diminish with the scaled distance (Figs. 10 and 11). On the one hand, the peak overpressure and maximum impulse decay rapidly with the standoff distance from the charge. On the other hand, the air flows from the high-pressure region to the low-pressure region as the shock wave propagates outward from the charge. Thus, the pressure field tends to be uniform.

Fig. 12 depicts the maximum shape factors ($S_{p,\max}$ and $S_{i,\max}$) for the overpressure and impulse versus the scaled distance for the initiation configurations at one end and at both ends. Compared to the results shown in Fig. 9, some important observations can be made from Fig. 12. The maximum shape factors ($S_{p,\max}$ and $S_{i,\max}$) generated from the single-end-initiated cylindrical charges [Figs. 12(a and b)] are generally larger than those generated from the other two initiation configurations. The maximum shape factors for the overpressure (impulse) can be as high as 4.5 (4.0). This is most likely attributed to the antisymmetric properties of this initiation configuration, by which more energy is directed along the gauge line $\theta = 0°$. As mentioned, if the cylindrical charge is center-initiated, the shock wave propagates faster in the axial than the radial direction at small scaled distances. Therefore, more energy is directed along the axial direction. If the cylindrical charge is

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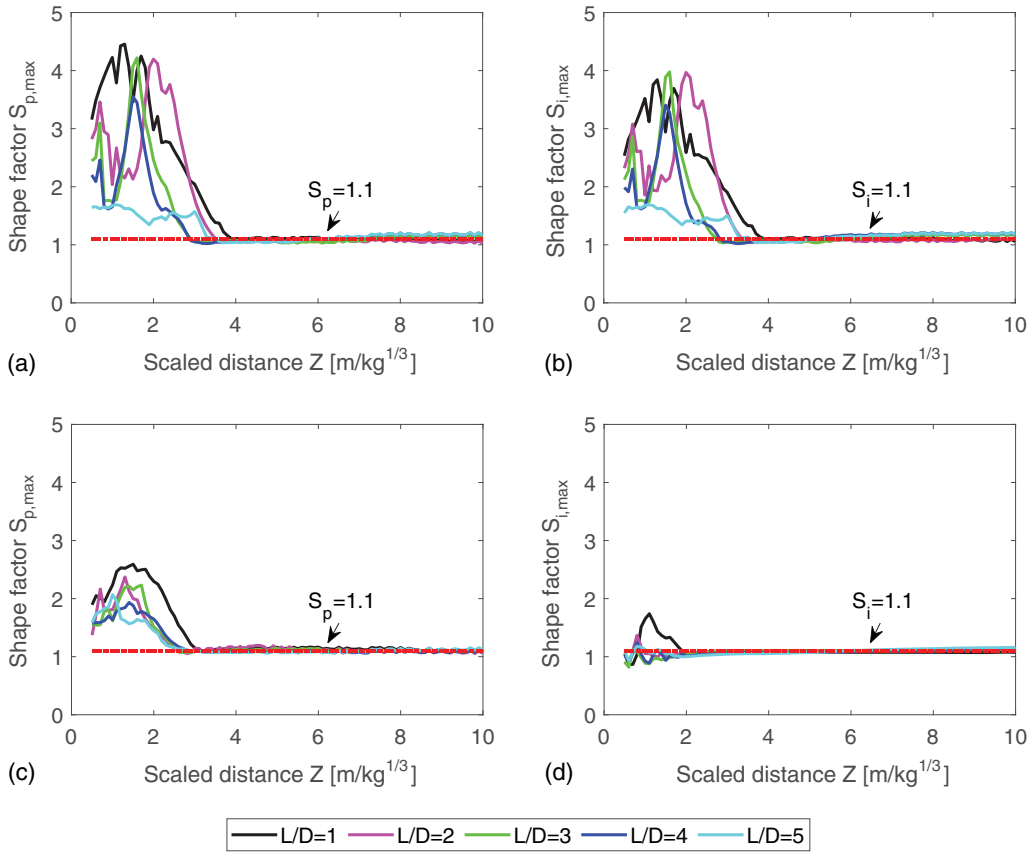



**Fig. 12.** Maximum shape factors ($S_{p,\max}$ and $S_{i,\max}$) for the overpressure and impulse versus the scaled distance dependent on the initiation configurations: (a) $S_{p,\max}$, single-end-initiated; (b) $S_{i,\max}$, single-end-initiated; (c) $S_{p,\max}$, double-end-initiated; and (d) $S_{i,\max}$, double-end-initiated.

initiated at both ends, more energy is directed along the radial direction compared to the center-initiated cylindrical charge, which compensates the difference of the directed energy between the axial and radial directions. This leads to slightly smaller values of the maximum shape factors $S_{p,\max}$ and $S_{i,\max}$ for the overpressure and impulse [Figs. 12(c and d)]. The single-end-initiated charge has the same influence range for the overpressure and impulse ($Z_p = Z_i = 3.9$ m/kg$^{1/3}$). The influence range for the overpressure of the two symmetric initiation configurations is more or less the same ($Z_p = 3.1$–$3.2$ m/kg$^{1/3}$), whereas the initiation configuration at both ends ($Z_i = 1.9$ m/kg$^{1/3}$) has a smaller influence range for the impulse than the initiation configuration at the center ($Z_i = 5.7$ m/kg$^{1/3}$). The initiation configuration at both ends is somewhat challenging for field experiments (e.g., Anastacio and Knock 2016; Hu et al. 2018; Chen et al. 2019) and extremely unlikely for weapons used by terrorists. However, one outcome from this study is that, if the blast loads are concerned, it is not the worst case of the three considered initiation configurations.

In summary, the blast loads (peak overpressure and maximum impulse) resulting from the three considered initiation configurations can be sorted in descending order, i.e., single-end-initiated > center-initiated > double-end-initiated. The initiation configuration at one end (at the center) has the largest influence range of $Z_p = 3.9$ m/kg$^{1/3}$ ($Z_i = 5.7$ m/kg$^{1/3}$) for the overpressure (impulse).

## Conclusions

This paper discussed the influence of cylindrical charge characteristics, i.e., the length-to-diameter ratio, azimuth angle, and initiation configuration, on blast loads (peak overpressure and maximum impulse). Numerical models were developed for the spherical and cylindrical charges. The length-to-diameter ratio varied from 1 to 5. Three different initiation configurations were considered, i.e., at the center, at one end, and at both ends. The numerical results were analyzed along the gauge lines around the cylindrical charges at the azimuth angles from 0° to 180°. A few conclusions are drawn from this numerical study:

- Regarding the centrally initiated cylindrical charges, the peak overpressure (maximum impulse) in the near field can be underestimated by a factor as high as 3.0 (1.9), if the charge shape effect is neglected. However, the shock front heals into a spherical one in the far field. This means that, for the blast-resistant design of protective structures, the charge shape should be directly considered in the near field to estimate the blast loads, whereas the blast loads generated from the spherical charge of the same mass can be used in the far field.
- The influence range beyond which the shape effect of centrally initiated cylindrical charges can be neglected is 5.7 m/kg$^{1/3}$ for the impulse, which is about twice that for the overpressure (3.2 m/kg$^{1/3}$).
- In general, the initiation configuration at one end generates the largest values of the blast loads among the three considered initiation configurations, which are as high as 4.5 (4.0) times the peak overpressure (maximum impulse) generated from the centrally initiated spherical charge of same mass.
- Roughly speaking, the maximum values of blast loads resulting from the three considered initiation configurations can be sorted in the descending order, i.e., at one end > at the center > at both ends.

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- The initiation configuration at one end (at the center) has the largest influence range of 3.9 $m/kg^{1/3}$ (5.7 $m/kg^{1/3}$) for the overpressure (impulse) among the three considered initiation configurations.

## Data Availability Statement

All data, models, or code generated or used during the study are available from the corresponding author by request.

## Acknowledgments

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